The standard uncertainty of a corrected measurement

$$u\left(Z_{ijk}^{*}\right) = \sqrt{u^2(Z_{ijk}) + u^2(\overline{CF_k})} \tag{8.13}$$

The relative uncertainty of a corrected measurement

$$UR\%\left(Z_{ijk}^{*}\right) = \frac{u\left(Z_{ijk}^{*}\right)}{Z_{ijk}^{*}} \tag{8.14}$$

As the standard uncertainty $u(Z_{ijk})$ is computed with the uncertainty function derived from the accuracy profile while the standard uncertainty of $\overline{CF_k}$ is given by formula (8.12), this procedure is easy to implement. New values of standard and relative uncertainty are obtained to readjust the uncertainty function. The latter will be applicable to any new sample analyzed and corrected by the laboratory.

But an additional question remains when using this method routinely. How to determine the correction factor to be applied to any new sample result? When the GUM correction approach is applied to each distinct level of the accuracy profile, it appears that the obtained correction factors and associated uncertainties vary with the concentration. If a new unknown sample is not close to one of the X_k value, an adjusted correction factor must be used.

The relationship between the concentration and the correction factor CF can be established and used to interpolate for any concentration. This must be preferably linear, since one of the validation requirements for analytical sciences, is that the trueness is proportional to the concentration, as explained in Section 4.1. Equation (8.15) is the mathematical interpretation of this requirement, where p is the proportionality coefficient. This is the equation of a straight-line forced through the origin because when the concentration is 0, no correction is required:

Proportionality relationship between the correction factors and the concentration

$$CF = p \times X \tag{8.15}$$

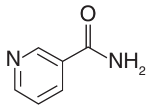
The coefficient p is the slope of the curve and can be positive or negative. In Equation (8.15), the predicted correction factor is expressed in the same units as X and can be applied directly to correct a value as in Equation (8.17). The next step is redrawing the initial MAP and obtaining a corrected profile. If it is satisfactory and the method validated, it can be considered as a verification of the relative correction factor as in equation (8.16). Another way consists in expressing the global correction factor as a percentage relative to the concentration. Then it applies to any recovery yield and makes it possible to compute the corrected accuracy profile.

Relative correction factor

$$\frac{CF}{X} \times 100 = p\% \tag{8.16}$$

Corrected measurement value

$$Z^{*} = Z \times (1 - p) \tag{8.17}$$

 Nicotinic acid	To illustrate this method, the NICOTINIC dataset described below is used. It is recognized in the food industry to supplement foods with diverse nutrients. The supplementation of cow milk with vitamin B3 (nicotinamide) or nicotinic acid is widely applied. Nicotinic acid is more stable and easily converted to nicotinamide in vivo. Nicotinamide is the biological form; it is a derivative of nicotinic acid and a water-soluble vitamin.

Although both have the same vitaminic activities, nicotinic acid is the preferred form for industrial food supplementation. The NICOTINIC method consisted of the determination of nicotinamide and nicotinic acid in cow's milk. The corresponding dataset is described in Table 8.5. Only the determination of nicotinic acid is addressed because the determination of nicotinamide was not questionable. Original inverse-predicted concentrations for the three validation materials are listed in Table 8.6. In the same table individual and average biases are reported.

As described in Section 5.2.2, the MAP from this data can be established when using the Resource H Excel worksheet, only considering β-ETIs. Results are illustrated in Figure 8.12 by the accuracy profile labeled "Before correction." There is an obvious and strong systematic relative bias close to −50% that requires being corrected. The main parameters of the MAP that confirm this graphical interpretation are listed below:

Concentration (mg/l)	0.2	2	4
Recovery yield (%)	53	54	52
Relative bias (%)	−47	−46	−48

The starting point of the correction procedure proposed by the GUM consists of calculating for each concentration level the average correction factor and its

Table 8.5 NICOTINIC – description of the dataset.

Title	**NICOTINIC**
Reference	Unpublished personal data
Measurand	Concentration of nicotinic acid in milk, expressed in mg/l.
Method	HPLC coupled to a fluorescence detector
Validation interval	[0.2, 4.0] mg/l
Acceptance interval	± 10%
Validation materials	3 validation materials containing 0.2, 2.0 and 4.0 μg/l respectively, prepared by SAM from a batch of homogenized milk.
Validation plan	Series ($I = 3$), replicates/series ($J = 3$), levels ($K = 3$)
Calibration plan	Series ($I' = 2$), replicates/series ($J' = 2$), levels ($K' = 2$)
Number of measures	12 measurements on calibration solutions 27 measurements on spiked materials.

Table 8.6 NICOTINIC – original inverse-predicted concentrations and bias in mg/l.

Level X_k	Series	Inverse predicted concentrations			Individual bias			Correction factor	
		Z_{i1k}	Z_{i2k}	Z_{i3k}	δ_{i1k}	δ_{i2k}	δ_{i3k}	$\overline{CF}_k$	$u^2(\overline{CF}_k) \times 10^{-3}$
0.2	1	0.096	0.113	0.109	−0.104	−0.087	−0.091	−0.093	0.598
	2	0.100	0.111	0.121	−0.100	−0.089	−0.079		
	3	0.100	0.105	0.105	−0.100	−0.095	−0.095		
2	1	1.167	1.090	1.043	−0.833	−0.910	−0.957	−0.922	5.084
	2	1.008	1.073	0.998	−0.992	−0.927	−1.002		
	3	1.141	1.004	1.181	−0.859	−0.996	−0.819		
4	1	2.196	1.933	2.087	−1.804	−2.067	−1.913	−1.934	11.839
	2	2.027	1.893	2.008	−1.973	−2.107	−1.992		
	3	2.171	2.178	2.102	−1.829	−1.822	−1.898		

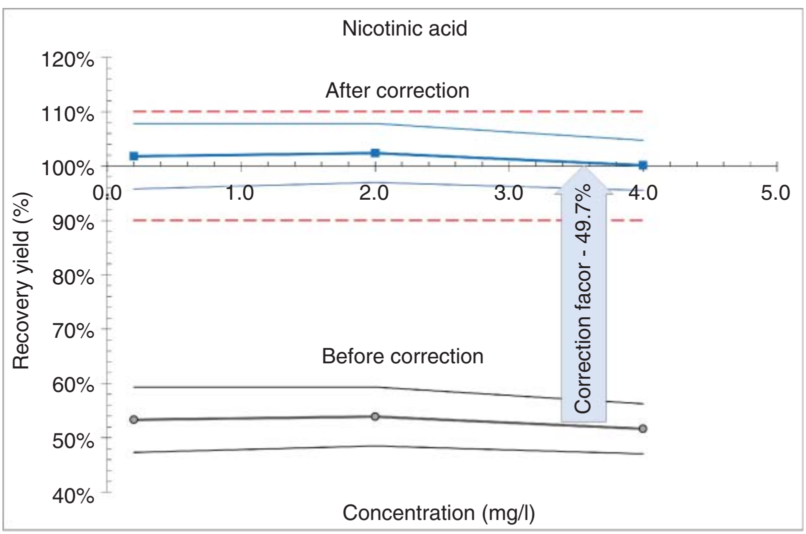


Figure 8.12 NICOTINIC – accuracy profiles before and after correction.

standard variance following Equations (8.10) and (8.12). The values obtained are reported in the last columns of Table 8.6. The most convenient way to compute the proportionality coefficient p is to apply the `LINEST` built-in function forcing the line to go through zero, i.e. setting the `Const` argument to `FALSE`. The value obtained is $p = -0.479$ expressed in mg/l; that gives $p\,\% = -47.9\%$ as a percentage.

While the initial accuracy profile is expressed as recovery yields, the corrected accuracy profile is obtained by subtracting this global correction factor from each value of the β-ETI bounds, themselves expressed in %. In Figure 8.12, the new

Table 8.7 NICOTINIC – influence of the correction factor on the uncertainty and accuracy profile.

	Parameters	Symbol	Concentration (mg/l) 0.2	2	4
			Accuracy profile		
Before	Recovery yield		53%	54%	52%
Before	Lower limit of β-ETI		47%	49%	47%
Before	Upper limit of β-ETI		59%	59%	56%
	Correction factor	$p\%$		−47.9%	
After	Recovery yield		101%	102%	100%
After	Lower limit of β-ETI		95%	96%	95%
After	Upper limit of β-ETI		107%	107%	104%
			Measurement uncertainty		
Before	Standard uncertainty	$u\,(Z)$	0.0086	0.0767	0.1249
Before	Relative uncertainty	$UR\,\%\,(Z)$	8.56%	7.67%	6.25%
Before	Standard variance	$u^2\,(Z)$	$7.32 \cdot 10^{-5}$	$5.88 \cdot 10^{-3}$	$1.56 \cdot 10^{-2}$
	Variance of the *CF*	$u^2(\overline{CF}_k)$	$5.98 \cdot 10^{-5}$	$5.08 \cdot 10^{-3}$	$1.19 \cdot 10^{-2}$
After	Standard uncertainty	$u(Z^*)$	0.01153	0.10472	0.16570
After	Relative uncertainty	$UR\,\%\,(Z^*)$	11.53%	10.47%	8.29%

accuracy profile is labeled "After correction." Table 8.7 table summarizes these calculations. Figure 8.12 perfectly illustrates how the global relative correction factor $p\%$ works, as the new corrected accuracy profile is now entirely within the acceptance interval limits.

Figure 8.13a shows the relationship between the correction factor and the concentration described by Equation (8.15). The linearity is established between the trueness and the concentration. However, the dispersion of individual biases around the average, symbolized by a small dashed line, increases when the concentration increases, and the WLS regression technique would improve the estimation of the slope of the straight-line. Figure 8.13a shows the relative uncertainty functions before and after correction.

These are power functions, as described in Section 7.5.2.

$$UR\%(Z) = 0.0753 \times Z^{-0.091}$$

$$UR\%(Z^*) = 0.1015 \times Z^{-0.093}$$

The function of the corrected measurements is shifted upwards, indicating that the uncertainty is positively impacted by the correction factor. This mainly affects the constant of the function, which increases from 0.0753 to 0.1015, i.e. a difference of about 0.025 that can be interpreted as 2.5% of increase in the relative uncertainty.

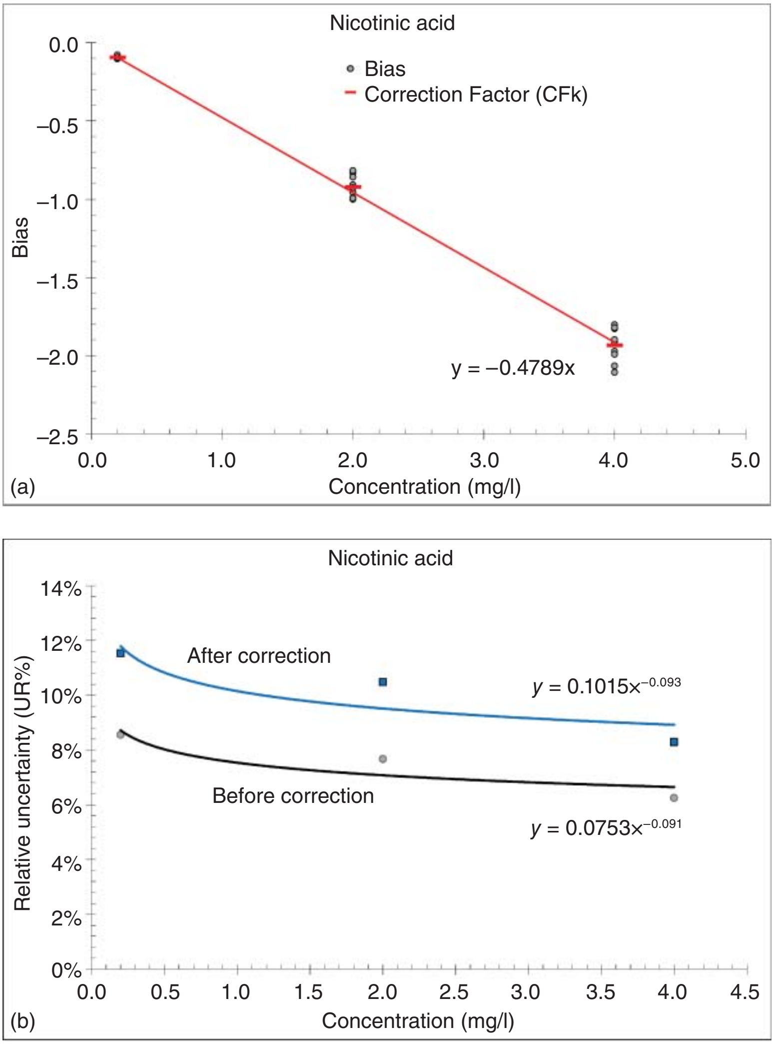


Figure 8.13 (a) NICOTINIC – relationship between the average correction factor and the concentration. (b) NICOTINIC –Relative uncertainty function before and after correction.

On the other hand, the power coefficients of both functions can be considered as constant. The reporting of the correction factor uncertainty only generates a constant shift. In conclusion, the correction of the measurement values increases the MU. In this example, the global correction factor is constant and close to 50%.

For example, let us take a milk sample for which the observed raw concentration of nicotinic acid Z=0.80 mg/l. After applying the correction factor $p = -0.479$, the corrected concentration becomes Z^{*}=0.80 × (1–(−0.479)) = 1.18 mg/l, and the relative uncertainty is estimated at 9.8% with a coverage interval of [1.07, 1.31].